

3(a)	<i>four from:</i> activity increases and decreases / activity peaks, for both ; A both have maximum activity / optimum temperature, at 60 °C activity is 76% at 40 °C for both ; (overall) activity decreases more steeply for free enzyme after, optimum / maximum / 60 °C ; A ora at 80 °C free enzyme is inactive / (fully) denatured / 0%, but immobilised enzyme, is still active / is <u>partially</u> denatured / 27% ; for immobilised enzyme, activity is higher (than free enzyme) at all temperatures above 60 °C ;	4
3(b)	<i>two from:</i> (alginate / immobilisation), is protective / has stabilising effect ; A enzyme, less / not, exposed (to solution) hydroxide ions do not penetrate the alginate beads ; shape of <u>active site</u> (of immobilised enzyme) is, less / not, disrupted / AW ; A active site is (more) complementary to substrate A few(er) active sites are, altered / changed few(er) bond(s) within (immobilised) enzyme break ; A hydrogen / ionic R peptide / disulfide at pH8 immobilised enzyme is not fully denatured ; A pH 7 / 7.5	2

Question	Answer	Marks
3(c)	<i>explanation must match variable stated</i> <i>standardised variable</i> concentration / volume, of, enzyme / substrate, solution ; <i>explanation for standardised variable</i> <i>idea that:</i> concentration / volume, of, enzyme / substrate, influences the probability of collisions between enzyme and substrate molecules / formation of enzyme/substrate complexes ; A number of / frequency of, collisions between substrate and enzyme or <i>standardised variable</i> use the same method for determining enzyme activity ; e.g. time when samples are taken <i>explanation for standardised variable 3</i> ref. to how enzyme activity is measured by conversion of substrate to product ;	2

Question	Answer	Marks
3(d)	<i>two from:</i> 1 (may be able to) obtain more product (per unit time) ; 2 can use higher temperatures (to obtain more product) / still active at higher temperatures ; A thermostable 3 does not denature (as easily as free) if temperatures increase / AW ; 4 immobilised enzyme can be reused / immobilised enzyme can be recovered ; 5 enzyme does not contaminate product ; A no effect on quality of product, e.g. taste 6 less, <u>purification</u> (of product) / downstream processing, needed ; A idea that downstream processing less, complex / difficult 7 longer shelf-life of enzyme / AW ; A durability 8 AVP ; e.g. allows continuous production (rather than batch) some enzymes are difficult to extract I ref. to cost / cost effective unqualified R reduces effect of end product inhibition	2